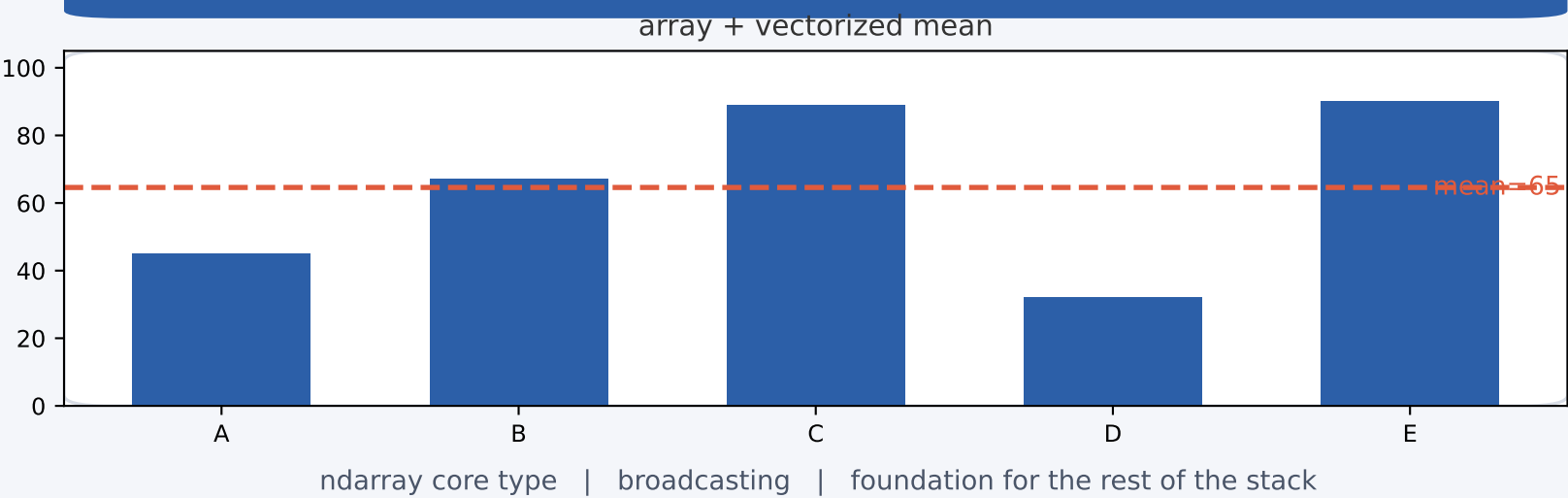


The Machine Learning Python Stack

Eight core libraries, what each one is for, and what its typical output looks like

NumPy

fast numerical arrays • since 2005



Pandas

tabular data & analysis • since 2008

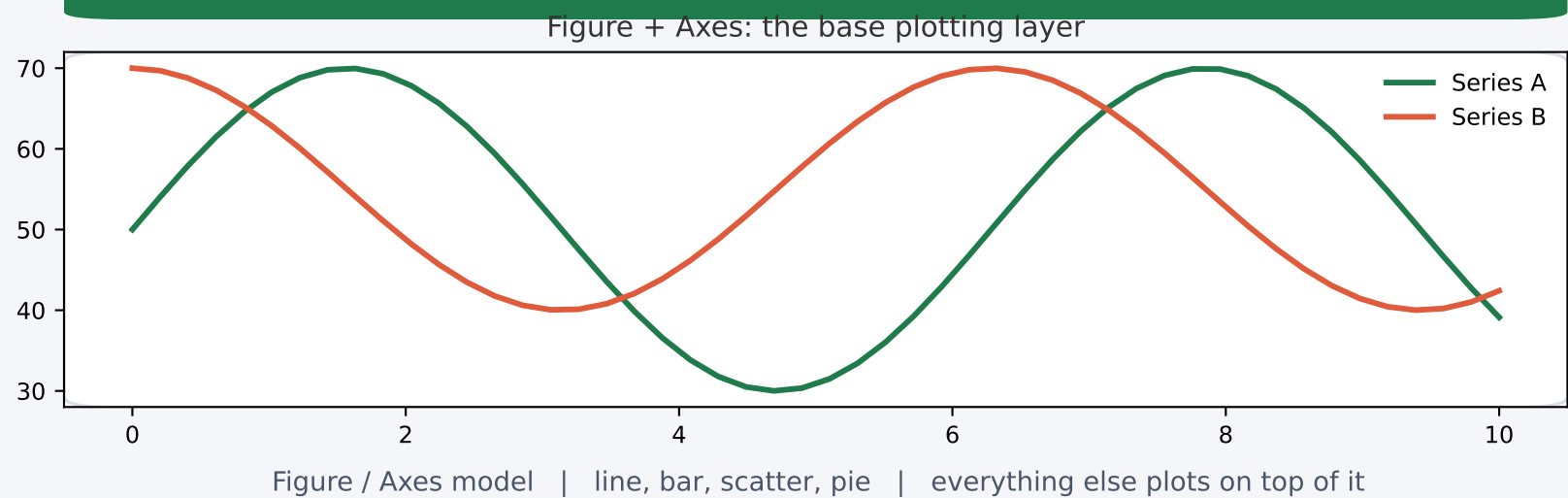
DataFrame: labeled rows & columns

Name	Math	Sci
Aryan	80	85
Bela	90	78
Chirag	45	50

Series / DataFrame | groupby, filter, sort | built on NumPy

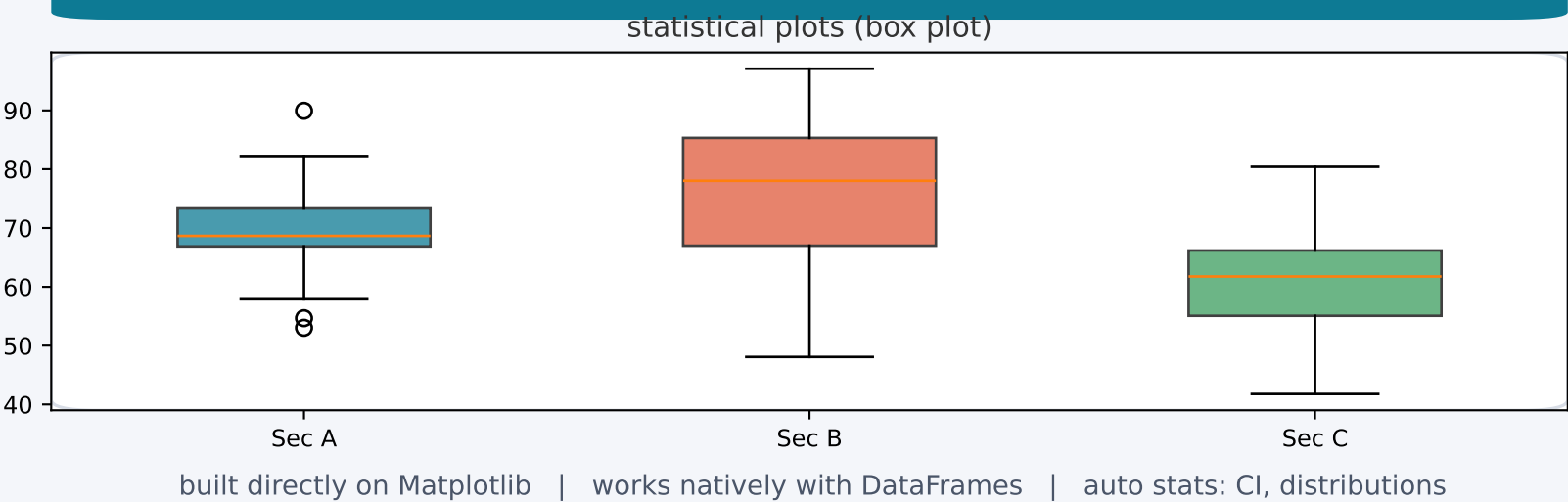
Matplotlib

the base plotting library • since 2003



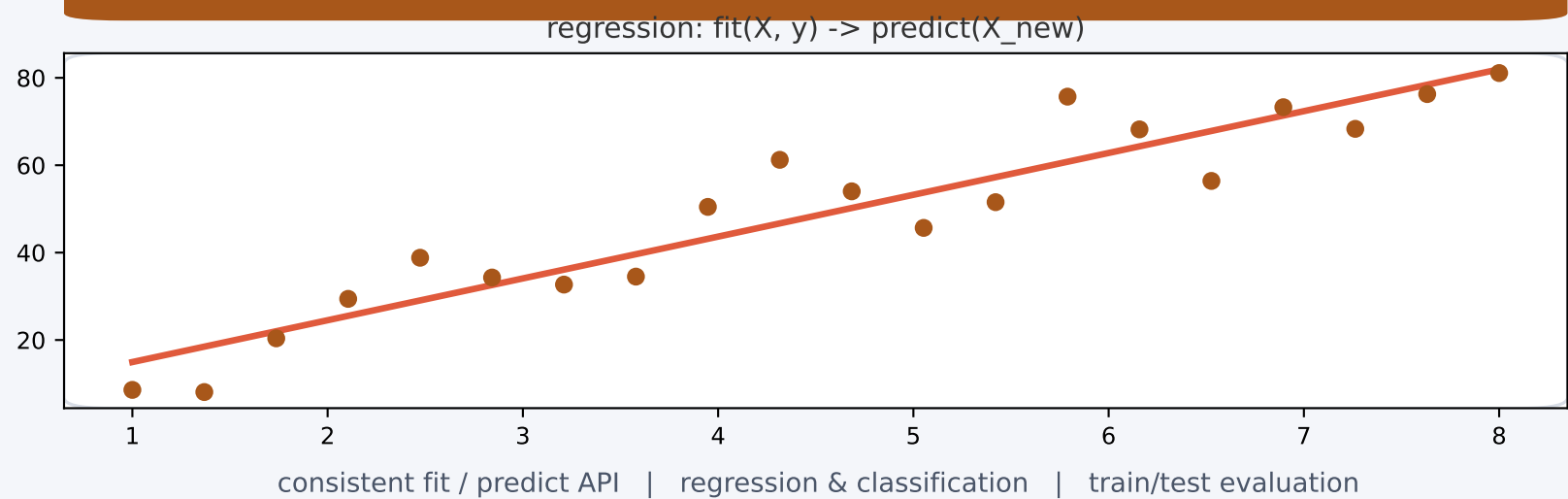
Seaborn

statistical visualization • since 2012



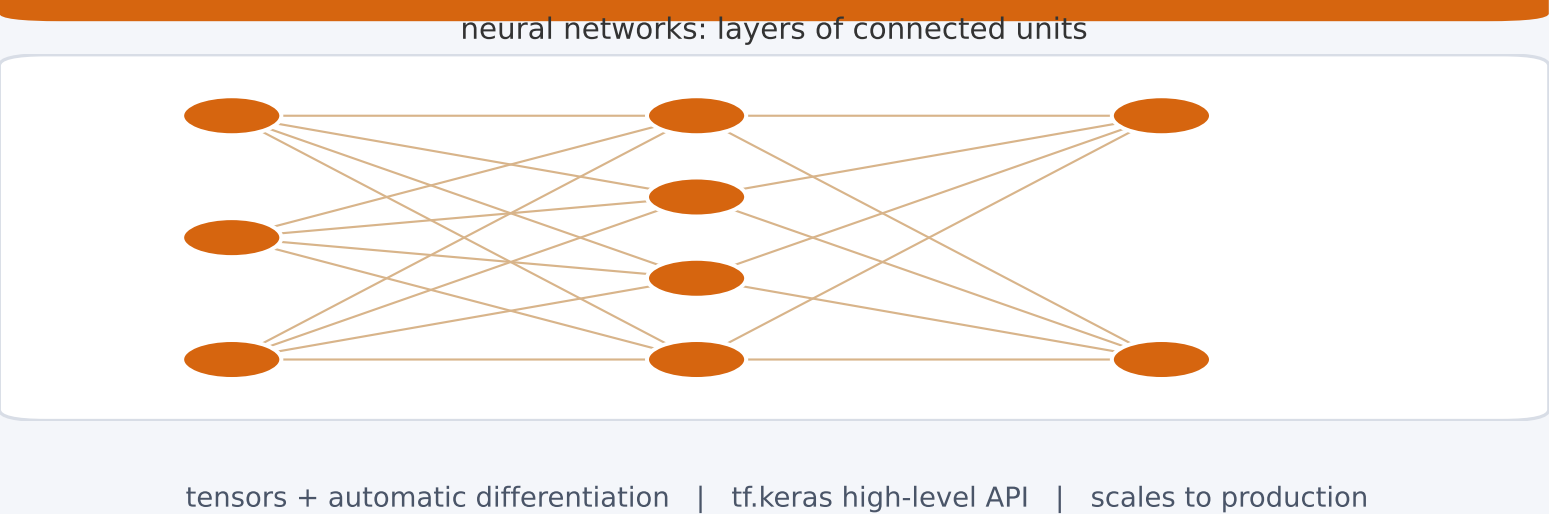
Scikit-learn

classical machine learning • since 2007



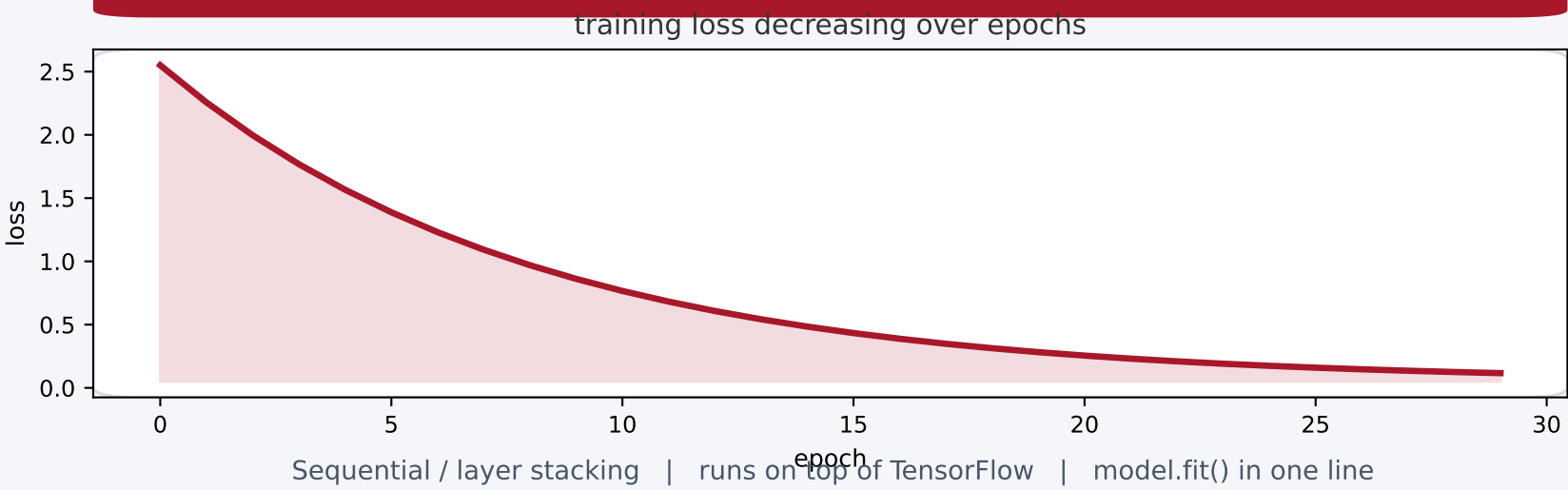
TensorFlow

deep learning (Google) • since 2015



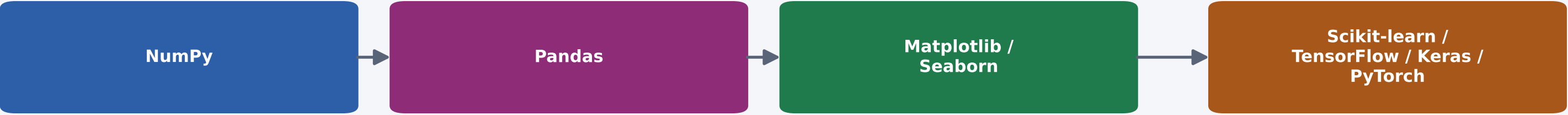
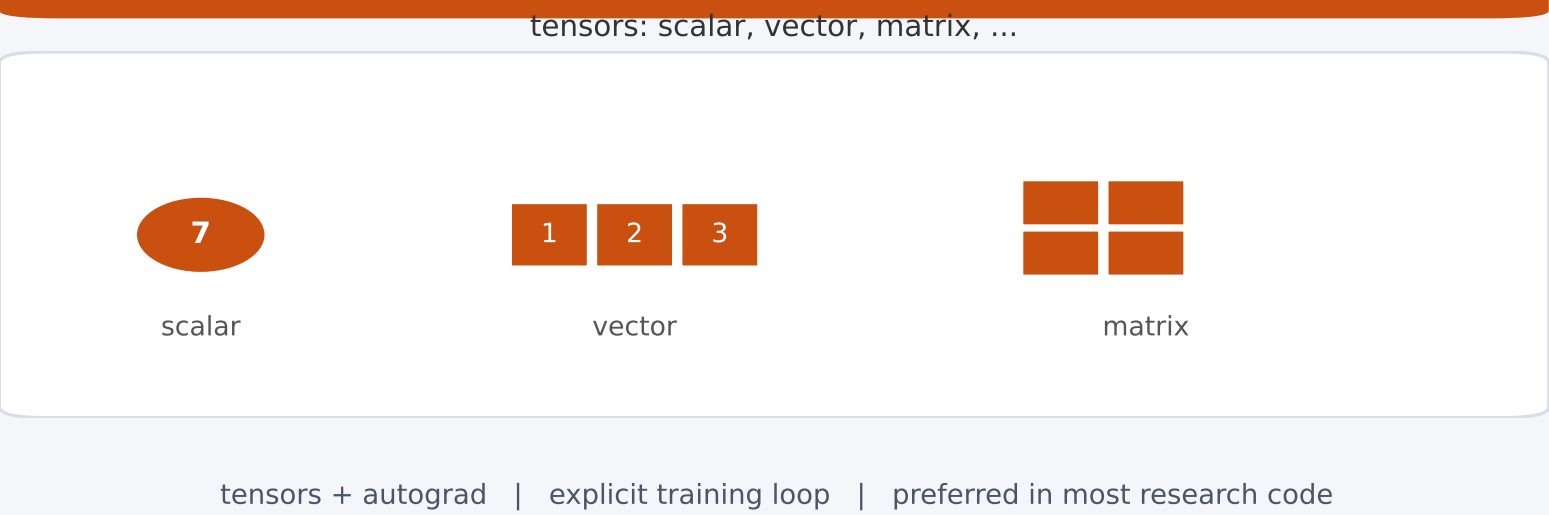
Keras

high-level deep learning API • since 2015



PyTorch

deep learning (Meta/FAIR) • since 2016



raw numbers -> organized data -> visual inspection -> modeling & prediction